

GeoNews — Week 35/2026

Weekly digest on Quaternary palaeoclimate research, East African rift dynamics, palaeohydrology and time-series analysis · 24 August 2026

What is GeoNews? An **automatically generated weekly digest**. A research workflow scans the publications of the past one to two weeks along with funding, policy and community sources, scores the hits against a fixed set of criteria, and assembles this issue from them.

The thematic profile is **aligned with the research interests of Markus L. Fischer** (Palaeoclimate Dynamics group, University of Potsdam): Quaternary palaeoclimate and palaeoenvironmental reconstruction, the East African Rift, the African Humid Period and human-environment coupling, lake-level and water-balance modelling, palaeofloods and fluvial geomorphology, ICDP drill cores, and linear and non-linear time-series analysis. Anyone working in these fields will probably find something useful here — it is explicitly not a complete survey of the Quaternary sciences.

Selection, scoring and short texts are produced by machine and reviewed before publication. Details may contain errors; in case of doubt the linked original source prevails.

Each paper receives a score from 0–100 across five criteria: thematic proximity (40), methodological proximity (25), impact/venue (15), connectivity (10) and recency (10). Categories: ● from 75 must-read · ● 50–74 relevant · ● 30–49 radar. The score measures proximity to this newsletter's profile and prioritises reading — it is not a quality judgement on the individual paper.

In brief

Eleven scored papers, and **no ● must-read** this week: no new publication in the window reaches the 75-point threshold. At the top are two ● papers. The first is a peer-reviewed *Climate of the Past* study of **deep-ocean density stratification across the Middle Pleistocene Transition (MPT)** from stacked Mg/Ca and benthic $\delta^{18}\text{O}$ series — one of the central reorganisations of the Quaternary climate system, touching the interplay of ocean stratification, the carbon cycle and ice volume. The second is a preprint on a **multidecadal monsoon drought in Seoul (1881–1907)**, reconstructed from one of the world's longest daily rain-gauge series and linked via model experiments to cool sea-surface conditions in the Indian Ocean and North Atlantic.

Behind them a broad ● radar list, led by a methodologically interesting preprint on **quantifying trends and transitions in a Miocene $\delta^{18}\text{O}$ stack** — deep-time in topic, but in its toolkit (detecting transitions in long proxy time series) directly connective to Quaternary tipping-point questions. Then palaeoceanographic and ice-core variability studies (Early Pliocene Arctic, East Antarctic snow accumulation, a Greenland fire proxy), a Pliocene NAO modelling study, and geomorphometric and fluvial-geomorphology preprints.

On how this issue came about — still constrained. Crossref access from the compute environment was blocked as in previous weeks (proxy 403); in addition, the browser-based query interface was unavailable in this automated run. The usual systematic ISSN query of the core journals via Crossref was therefore not possible. Instead, the review ran via web search plus the

freely accessible, dated "recent" lists of the Copernicus journals (*Climate of the Past*, *Earth Surface Dynamics*, EGU sphere preprints); for every scored entry the short summary/abstract there was consulted. This shifts coverage toward open-access titles. Elsevier and Wiley core journals (QSR, EPSL, GPC, P³, Boreas, JQS) could not be reliably captured with title and date without Crossref and are therefore **not** carried as a separate block (rather than inventing results). The dates given are journal publication dates, not Crossref created dates. A direct East Africa / rift / AHP bullseye from an African primary study was not findable within the window.

On funding, the focus is unchanged: the **PAGES applications** (deadline **11 Sep 2026**, in just over two and a half weeks), the **ERC Starting Grant 2027 (14 Oct 2026)** and the Horizon palaeoclimate call **HORIZON-CL5-2027-01-D1-08** (opens 17 Nov 2026, deadline 4 Mar 2027). In science policy, the **WissZeitVG amendment** adopted by the federal cabinet on 29 Jul 2026 remains the dominant topic; the Bundestag debate is expected after the summer recess in September.

Part A — Papers of the week

Captured were papers with a publication date from 12 August 2026 onward that were not already listed in an earlier issue. DOIs already listed are excluded. All scored entries rely on a consulted short summary/abstract. A block of Elsevier/Wiley titles without abstract is omitted this week because the Crossref query needed for it was unavailable (see methods).

● **Must-read (from 75)**

No paper in this category this week: no new publication in the window reaches 75 points.

● **Relevant (50-74)**

1. Increased abyssal ocean density stratification across the Middle Pleistocene Transition · *Climate of the Past*, 2026 · **Score 69** A reconstruction of interbasinal deep-sea temperature and salinity gradients over the past 1.5 million years, based on stacked Mg/Ca and benthic $\delta^{18}\text{O}$ datasets. Result: across the Middle Pleistocene Transition (MPT) the deep Atlantic water mass cooled while the Pacific became more saline — deep-ocean density stratification increased. The authors argue that the glacial ocean thereby became a more effective carbon trap, which lowered atmospheric pCO_2 and favoured the growth of larger ice sheets; they read this as a physical contribution to explaining the MPT. Connective as a Quaternary palaeoceanography study (Tier 2): the MPT is one of the central reorganisations of the Quaternary climate system, and the question of how ocean stratification, the carbon cycle and ice volume interact touches the same forcing decomposition that is also central to continental water-balance and monsoon dynamics across the glacial-interglacial cycle. Methodologically a clean example of stacking and interbasinal differencing of long proxy time series. *Theme 25* · *Method 17* · *Venue 11* · *Connectivity 6* · *Recency 10* (publ. 19 Aug 2026) · DOI: [10.5194/cp-22-1559-2026](https://doi.org/10.5194/cp-22-1559-2026)

2. Sustained Late-Nineteenth-Century Monsoon Drought in Seoul and the Indian Ocean-North Atlantic Forcing of Its Circulation Weakening · *EGU sphere* (preprint, under review for *Climate of the Past*), 2026 · **Score 60** An analysis of one of the world's longest daily precipitation series: the Korean royal chronicles preserve rain-gauge readings for Seoul back to 1777. After checking doubtful entries against the original manuscripts, the study documents a sustained drought — summer rainfall fell by about a quarter over 27 years (1881–1907). Climate model experiments, per the abstract, trace the cause to a weakened monsoon circulation driven

mainly by unusually cool waters in the Indian Ocean and North Atlantic. Connective as a palaeohydrological / climate-dynamics study (Tier 2): multidecadal rainfall persistence and its remote coupling to oceanic boundary conditions are methodologically the same attribution question about long-lasting dry phases that is central in continental hydroclimate archives — here through a documentary instrumental series rather than a sediment archive. Preprint, hence the low venue weight; the finding stands provisionally pending completion of review. *Theme 25 · Method 15 · Venue 4 · Connectivity 6 · Recency 10* (publ. 18 Aug 2026) · DOI: [10.5194/egusphere-2026-4812](https://doi.org/10.5194/egusphere-2026-4812)

● Radar

- **Quantifying trends and transitions in a Miocene benthic $\delta^{18}\text{O}$ stack** · *EGUsphere* (preprint) · **48** ● · Benthic $\delta^{18}\text{O}$ stratigraphy to correlate Miocene records in time and to quantify global trends and transitions (ice volume / deep-sea temperature). Deep-time in topic and therefore at the edge of the profile, but methodologically noteworthy: the statistical detection of trends and transitions in a long proxy time series transfers directly to Quaternary tipping-point and transition questions. DOI: [10.5194/egusphere-2026-4648](https://doi.org/10.5194/egusphere-2026-4648) · publ. 12 Aug 2026
- **Enhanced ocean mixing improves proxy-model agreement for an Early Pliocene seasonally ice-free Arctic Ocean** · *EGUsphere* (preprint) · **43** ● · A proxy-model comparison for a seasonally ice-free Arctic Ocean in the Early Pliocene; enhanced ocean mixing improves agreement with the proxy evidence. Noted as a warm-climate palaeoceanography modelling study at the edge of the profile. DOI: [10.5194/egusphere-2026-4723](https://doi.org/10.5194/egusphere-2026-4723) · publ. 18 Aug 2026
- **High centennial-scale variability in an East Antarctic snow accumulation record spanning the last millennium** · *EGUsphere* (preprint) · **43** ● · A reconstruction of annual snow accumulation from the Mount Brown South ice core over the last millennium; several century-long phases with markedly less accumulation than in the satellite era. Noted as an ice-core variability study at the edge of the profile. DOI: [10.5194/egusphere-2026-4880](https://doi.org/10.5194/egusphere-2026-4880) · publ. 18 Aug 2026
- **The scale-mosaic hypothesis: a framework for heterogeneous scale selection in geomorphometry and DEM-based terrain analysis** · *EGUsphere* (preprint) · **43** ● · Locally optimised scale selection per grid cell instead of a single scale for DEM-based terrain analysis. Noted as a geomorphometric-methodological study (Tier 3, GIS/remote sensing). DOI: [10.5194/egusphere-2026-4402](https://doi.org/10.5194/egusphere-2026-4402) · publ. 21 Aug 2026
- **Oleamide as a novel insoluble organic marker for reconstructing Arctic wildfire history in the ice core from southeastern Greenland** · *EGUsphere* (preprint) · **40** ● · First detection of water-insoluble oleamide in a Greenland ice core (1799–2020) as a new, immobile fire proxy; its peaks match major wildfires in 1908, 1961 and 1994. Noted as a proxy-method study at the edge of the profile. DOI: [10.5194/egusphere-2026-4810](https://doi.org/10.5194/egusphere-2026-4810) · publ. 14 Aug 2026
- **An assessment of the North Atlantic Oscillation in the Late Pliocene and its impact on European climate** · *EGUsphere* (preprint) · **38** ● · A multi-model analysis of the NAO in the Late Pliocene; a more positive NAO index on average with a large model spread, and a tendency toward more precipitation in Northern Europe. Noted as a climate-dynamics modelling study at the edge of the profile. DOI: [10.5194/egusphere-2026-4616](https://doi.org/10.5194/egusphere-2026-4616) · publ. 12 Aug 2026
- **Bank Pull or Bar Push: Who Leads the Dance of Meander Migration?** · *EGUsphere* (preprint) · **38** ● · Almost four decades of satellite observation of 4,923 river bends worldwide; meander migration begins with either erosion or deposition depending on environmental

conditions. Noted as a fluvial-geomorphology remote-sensing study (Tier 3). DOI: [10.5194/egusphere-2026-4621](https://doi.org/10.5194/egusphere-2026-4621) · publ. 12 Aug 2026

- **Coarsening of fluvial sediments after the Gorkha Earthquake 2015, Nepal** · *EGUsphere* (preprint) · **37** ● · Coarser suspended sediment in a Himalayan river basin during the first year after the Gorkha earthquake, with the signal fading downstream. Noted as a fluvial-geomorphology study (Tier 3) of earthquake-induced sediment dynamics. DOI: [10.5194/egusphere-2026-4202](https://doi.org/10.5194/egusphere-2026-4202) · publ. 14 Aug 2026
- **Wine, victuals and volcanoes - potential impacts of the 1640/1641 double volcanic eruption on climate and society in early modern Switzerland** · *EGUsphere* (preprint) · **36** ● · Archival material and reanalysis data on the consequences of the 1640/41 double eruption for agriculture and society on the Swiss Plateau. Noted as a historical-climatology study (Tier 3) at the edge of the profile. DOI: [10.5194/egusphere-2026-4370](https://doi.org/10.5194/egusphere-2026-4370) · publ. 12 Aug 2026

Part B — Funding & calls

PAGES — working-group and workshop support, deadline 11 Sep 2026. Next application round for workshops, meetings, new working groups and extensions of existing groups — in just over two and a half weeks. Applicants should contact an SSC member at least two weeks before the deadline, and workshop support should be requested at least six months before the event. Ahead of the Horizon palaeoclimate call, a convenient instrument for forming a consortium visibly. In addition: **PAGES Early-Career Award 2027**, deadline **29 Sep 2026**. Sources: [PAGES — Workshop/meeting support](#) · [PAGES Newsletter July 2026](#).

ERC Starting Grant 2027 — deadline 14 Oct 2026. The call (ERC-2027-STG) has been open since 22 Jul 2026. Eligibility: PhD defence between 1 Jan 2017 and 31 Dec 2026 (up to ten years before the reference year), extendable for parental and care leave. Funding up to EUR 1.5 million over five years (plus up to EUR 1 million for start-up/equipment). Step-1 results in April 2027, interviews and Step-2 results in August 2027. New from the 2027 calls: at most one Starting and one Consolidator Grant per career. The **ERC Consolidator 2027** is expected for January 2027. Sources: [ERC — Starting Grant](#) · [ERC — Applying in the 2027 competitions](#).

Horizon Europe — "Palaeoclimate science for a better understanding of Earth system dynamics" (HORIZON-CL5-2027-01-D1-08). Opens **17 Nov 2026**, deadline **4 Mar 2027**, Research and Innovation Action, budget about **EUR 14 million for roughly two projects**, i.e. some EUR 7 million per consortium. Aim per the call text: robust information on palaeoclimate states and events outside the variability of recent centuries, as a basis for advancing Earth system models. With only two projects funded Europe-wide the competition is fierce; consortium talks should be started early. Sources: [Topic page](#) · [CL5 Work Programme 2026–2027 \(PDF\)](#).

ICDP — proposal cycle. The regular annual deadline for full, pre- and workshop proposals is **15 January**; the 2027 cycle is likely to follow the same pattern, though no formal announcement has appeared. Workshop calls run through the year. Source: [ICDP — Proposals](#).

DFG — changed framework conditions. No new palaeoclimate-specific call. Important for planning: the **programme allowance rises from 22 to 25 percent from 2027**; the next **Priority Programme application round** will only take place at the **end of 2026**, with funding starting from **2028**. Additionally, at the margin of subject funding: applications for the **DFG**

Communicator Award 2027 (science communication) remain open until **30 Sep 2026**.

Sources: [Forschung & Lehre — 25 instead of 22 percent from 2027](#) · [DFG — Press releases 2026](#).

Part C — Science policy (DE/EU)

WissZeitVG — amendment adopted by the cabinet on 29 Jul 2026, Bundestag to follow in the autumn. The federal cabinet adopted the bill to modernise fixed-term employment law in science on 29 July 2026; the May draft was revised in parts after criticism from associations. At its core are, for the first time, binding **minimum contract durations: three years for doctoral researchers' first contracts, two years for postdocs**; the maximum fixed-term duration in the postdoc phase is **standardised at six years**. For **student research assistants** the draft provides a minimum term of one year and raises the maximum fixed-term duration from six to eight years. According to parliamentary sources, the Bundestag debate is to be expected after the summer recess, i.e. in September. As interpretation, not established fact: minimum durations improve the predictability of early contracts, while the removal of earlier carry-over flexibility further narrows individual fixed-term planning in the postdoc phase — the further parliamentary path remains to be watched. Sources: [BMFTR — Cabinet adopts amendment to the WissZeitVG](#) · [Bundestag — Reform of the Wissenschaftszeitvertragsgesetz](#) · [Wiarda blog — WissZeitVG: BMFTR presents draft](#).

Federal budget 2027 — cabinet decision of 6 Jul 2026, Bundestag to follow in the autumn. The government draft has stood since 6 July 2026. The BMFTR core budget is about **EUR 22 billion** — a nominal increase of only about **0.7 percent** over 2026. Real increases are almost exclusively in the special fund for infrastructure and climate neutrality, whose R&D investments rise markedly in 2027 over 2026. The first parliamentary reading is due in the autumn. As interpretation: the regular budget stagnates in real terms given current pay and materials cost rises, keeping the operating situation of the funding organisations under pressure. Sources: [Forschung & Lehre — Only a slim increase for the BMFTR core budget](#) · [Table.Briefings — Budget 2027](#).

DFG cuts at the postdoc level remain in force. Owing to high pay rises and increased equipment and consumables costs that the promised annual funding increments do not cover, the DFG has postponed the funding start of new Research Training Group applications and **suspended the approval of postdoc positions**. For postdoc career paths in the geosciences this removes a funding route for now, increasing the pressure on individual-grant and EU funds. Source: [Fakultätentag Informatik](#).

Part D — Community & dates

- **PAGES working-group and workshop applications — 11 Sep 2026** (in just over two and a half weeks) · **PAGES Early-Career Award 2027 — 29 Sep 2026**.
- **DEUQUA meeting 2026 — 28 Sep to 2 Oct 2026 in Freiburg im Breisgau**, theme "Environmental Records and Landscape Change", with full-day excursions on 28 Sep and 2 Oct (Black Forest/Vosges glaciation, Rhine Valley loess, fluvial dynamics in the Upper Rhine Graben, among others). Source: [DEUQUA 2026](#).
- **DFG Communicator Award 2027 — deadline 30 Sep 2026**.

- **ERC Starting Grant 2027** — deadline **14 Oct 2026**.
- **Horizon CL5 palaeoclimate call** — opens **17 Nov 2026**, deadline **4 Mar 2027**.
- **AGU Fall Meeting 2026** — **7–11 Dec 2026 in San Francisco**; the abstract deadline (5 Aug 2026) has passed. Source: [AGU](#).
- **EGU General Assembly 2027** — **4 to 9 Apr 2027 in Vienna**, hybrid. Abstract deadline not yet published; typically mid-January. Source: [EGU](#).
- **ICDP** — next regular proposal deadline expected **15 Jan 2027**, following previous years and not yet confirmed.

Methods & sources

The paper review follows the fixed set of criteria (thematic proximity, methodological proximity, impact/venue, connectivity, recency) and excludes DOIs already listed in an earlier issue. Funding opportunities, science policy and dates rely on the sources linked in Parts B to D.

On how this issue came about — a different research path. Crossref access from the compute environment was blocked as in previous weeks (proxy 403); in addition, the browser-based query interface otherwise used as a fallback was unavailable in this automated run. The systematic ISSN query of the core journals via the Crossref REST API could therefore not run. The review instead used web search combined with the freely accessible, dated "recent" lists and article landing pages of the Copernicus journals — in particular *Climate of the Past*, *Earth Surface Dynamics* and the EGU sphere preprints. For every scored entry the short summary or abstract was consulted on the respective landing page.

What this means for coverage. The selection is shifted this week toward freely accessible titles because their abstracts are directly readable. The Elsevier (QSR, EPSL, GPC, P³, Catena, Geomorphology) and Wiley core journals (Boreas, JQS) could not be reliably captured with title and date within the window without Crossref; a title-only block without abstracts is therefore omitted rather than deriving results from incomplete hit lists. The dates given are **journal publication dates** (landing page), not Crossref `created` dates; the two usually differ by a few days. A direct East Africa / rift / AHP bullseye from an African primary study was not findable within the window; the profile is served this week through Quaternary palaeoceanography (MPT), monsoon/hydroclimate dynamics (Seoul) and the methods side (time-series transition detection). For context, several thematically fitting Nature-family papers falling outside the window were checked and excluded (among others a Sahara AHP drought study, publ. 25 Mar 2026; a Makgadikgadi lake-level record, publ. 18 May 2026; a Tunisian speleothem record, publ. 10 Feb 2026).

GeoNews is an automatically generated digest, published by Dr Markus L. Fischer, Palaeoclimate Dynamics group, University of Potsdam. The thematic profile follows his research interests; selection is made by a rule-based workflow, not an editorial team. The digest is an independent literature overview with no connection to the publishers or funding bodies named, and constitutes no recommendation or advice. Corrections and tips are welcome.